

Abstract

This report documents the sensing, monitoring, and control system developed by our group for a desiccant-enhanced evaporative water cooler prototype. While a conventional direct evaporative cooler is limited in humid climates by a reduced wet-bulb depression, the desiccant hybrid system overcomes this by first stripping moisture from ambient air before passing it through evaporative pads. Our group was assigned the responsibility of designing, procuring, and integrating the complete electronic monitoring and automation layer of this system.

The scope of our work encompasses the selection and installation of four distinct sensing subsystems — temperature and humidity sensing via DHT22 transducers at three process points, differential pressure monitoring via an MPX5010DP sensor for real-time desiccant saturation assessment, and water level protection via a float switch — all interfaced to an Arduino Uno microcontroller. The Arduino processes real-time sensor inputs, computes performance metrics (temperature reduction, desiccant effect, pressure drop), and autonomously controls the submersible water pump through a relay module using a hysteresis-based temperature control algorithm. All sensor readings and computed metrics are output to a Serial Monitor at 9600 baud during laboratory demonstration. Additionally, a YF-S201 flow sensor monitors water delivery to the evaporative pad.

A complete procurement analysis was conducted, and all sensor mounting hardware and fittings were sourced from local markets. The total investment for our group's scope amounts to approximately 7,480 PKR for electronic components and 1,340 PKR for mounting fittings and hardware, giving a combined group expenditure of approximately 8,820 PKR.

Group Details & Contributions

This section presents the official roster and task distribution for the 7-member team responsible for sensor integration, Arduino control logic, and electronic monitoring of the desiccant-enhanced water cooler project.

Sr. No.	Student Name	Reg. No.	Core Project Responsibility
1	Fahad Fayyaz	2024-CD-01	Arduino Code Debugging
2	Mian M. Azib Atif	2024-ME-52	Sensor Wiring & Circuit Assembly
3	Hashir Noor	2024-ME-53	Cost Analysis & Documentation
4	Umar Farooq	2024-ME-56	Procurement Support
5	Haroon Rizwan	2024-ME-65	Sensor Integration Lead & Arduino Programming
6	Abdullah Butt	2024-ME-69	Sensor Mounting & Physical Fabrication
7	Ramish Masood	2024-ME-77	System Testing, Verification & Calibration

1. Introduction & Project Overview

1.1 Project Background

Traditional evaporative cooling systems — commonly known as desert coolers — operate on the principle of latent heat of vaporization. Hot, dry air is drawn through water-saturated pads, and as water evaporates into the air stream, it absorbs heat, thereby reducing the dry-bulb temperature of the air. While this approach is effective in arid regions, its thermodynamic performance degrades substantially in humid environments. As ambient relative humidity rises, the air's capacity to accept additional moisture diminishes, and the wet-bulb depression — the maximum achievable temperature drop — shrinks to near zero. The cooler then produces air that is both insufficiently cool and uncomfortably humid.

To overcome this limitation, this project introduces a Desiccant-Enhanced Evaporative Water Cooler that separates the conditioning process into two thermodynamic stages:

1. **Dehumidification (Adsorption):** Ambient air is first passed through a rotating desiccant wheel coated with silica gel or molecular sieve. The desiccant material adsorbs moisture from the air via vapor pressure differentials, producing a hot, dry air stream.
2. **Sensible & Latent Cooling:** The pre-conditioned, low-humidity air is then routed through high-density cross-fluted cellulose honeycomb evaporative pads. With very low initial humidity, the evaporation rate is high, achieving a substantial temperature drop without saturating the conditioned space.

Continuous operation requires a regeneration loop: a heating element and high-speed auxiliary fan drive moisture out of the saturated desiccant disc and exhaust it to the atmosphere, restoring the disc's adsorptive capacity. The entire system relies on coordinated operation of fans, pump, and heater — which is where our group's responsibility begins.

1.2 Scope of Responsibility (Group: Sensing, Monitoring & Control)

Whereas other groups focused on structural fabrication, casing manufacture, hydraulic circuit assembly, and desiccant chemistry, our group was tasked with designing and implementing the system's intelligence layer — the sensing, monitoring, and automated control subsystem.

Our group's deliverables span three core operational areas:

- **Sensor Procurement & Installation:** Selecting, sourcing, and physically mounting all sensing transducers at correct measurement points within the cooler prototype.
- **Arduino-Based Control Logic:** Programming the Arduino Uno to read sensor data, compute real-time performance metrics, and autonomously control the pump relay based on temperature hysteresis and water level status.
- **Serial Monitoring & Verification:** Outputting all live readings and performance metrics to the Serial Monitor at 9600 baud during laboratory demonstration, allowing the instructor to verify that the system is actively dehumidifying and cooling.

2. Detailed Sensor & Component Specifications

This section outlines the precise specifications of every sensing and control component procured and installed by our group, along with the engineering rationale for each selection.

2.1 DHT22 Temperature & Humidity Sensor (×3)

- **Model / Identifier:**DHT22 (AM2302)
- **Measurement Range:**Temperature: -40°C to $+80^{\circ}\text{C}$ | Relative Humidity: 0% to 100% RH
- **Accuracy:** $\pm 0.5^{\circ}\text{C}$ temperature, $\pm 2-5\%$ RH
- **Interface:**Single-wire digital signal; requires $4.7\text{k}\Omega$ – $10\text{k}\Omega$ pull-up resistor on data line
- **Operating Voltage:** 3.3 – 5V DC
- **Arduino Pin Assignment:**DHT22 #1 → D4 (Inlet) | DHT22 #2 → D7 (Post-Desiccant) | DHT22 #3 → D8 (Outlet)

Engineering Purpose:

The DHT22 is the primary thermodynamic measurement instrument in this system. Three units are deployed at distinct process points: one at the system air inlet (measures ambient conditions entering the desiccant wheel), one after the desiccant wheel (confirms that humidity has been reduced and temperature has risen due to adsorption heat), and one at the final cooled outlet (proves that the evaporative stage has reduced dry-bulb temperature). The Arduino uses these three readings to compute the desiccant effect (inlet RH minus post-desiccant RH) and overall temperature reduction (inlet temp minus outlet temp) — the core thermodynamic evidence required for this OEL. The DHT22 was chosen over the simpler DHT11 for its superior accuracy and wider temperature range, using the Adafruit DHT sensor library.

2.2 DS18B20 Digital Temperature Sensor (×2)

- **Model / Identifier:**DS18B20 Waterproof Probe
- **Measurement Range:** -55°C to $+125^{\circ}\text{C}$
- **Accuracy:** $\pm 0.5^{\circ}\text{C}$ (from -10°C to $+85^{\circ}\text{C}$)
- **Interface:**1-Wire protocol; requires a single $4.7\text{k}\Omega$ pull-up resistor
- **Special Feature:**Stainless steel waterproof probe — survives high-temperature and high-humidity environments

Engineering Purpose:

The desiccant regeneration zone reaches elevated temperatures due to the embedded heating element used to drive adsorbed moisture out of the desiccant disc. A DHT22 would be damaged or produce erratic readings at these temperatures. The DS18B20's rated operating range of up to 125°C and its robust stainless-steel probe construction make it the appropriate choice for this zone. Two units are deployed: one on the inlet side of the heating element and one on the exhaust duct, allowing monitoring of regeneration zone temperatures to confirm effective desiccant drying.

2.3 MPX5010DP Differential Pressure Sensor (×1)

- **Model / Identifier:**Freescale / NXP MPX5010DP
- **Pressure Range:**0 to 10 kPa differential
- **Output:**Ratiometric analog voltage (0.2V – 4.7V)
- **Interface:**Analog output → Arduino pin A0 (10-bit ADC); two barbed pressure ports for tubing

- **Supply Voltage:**5V DC
- **Transfer Function:** $V_{out} = V_s \times (0.09 \times P + 0.04)$ where P is in kPa and $V_s = 5V$

Engineering Purpose:

As the desiccant disc adsorbs moisture and becomes progressively saturated, its resistance to airflow increases. This manifests as a rising pressure drop across the disc. The MPX5010DP continuously monitors this differential pressure via its analog output, which the Arduino converts to kPa using the datasheet transfer function implemented in the `readPressure_kPa()` function. The computed pressure value is reported to the Serial Monitor in real time. The aquarium air tubes and Teflon-sealed fittings are used to connect this sensor's ports to the measurement taps on either side of the desiccant disc.

2.4 Float Switch — Water Level Sensor (×1)

- **Type:**Mechanical magnetic float switch (normally open)
- **Operating Voltage:**Up to 24V DC / AC
- **Arduino Pin:**D6 (configured as `INPUT_PULLUP`)
- **Logic:**LOW = water present (float closed) | HIGH = tank empty (float open)

Engineering Purpose:

The submersible centrifugal water pump must not run dry, as operating without water causes motor burnout within minutes. The float switch is installed inside the water reservoir at the minimum safe water level. When water drops below this level, the float falls and opens the circuit; the Arduino reads HIGH on D6 and immediately forces the pump relay OFF, printing a TANK EMPTY warning to Serial. This protective function takes absolute priority over temperature-based pump control in the code logic.

2.5 Arduino Uno Microcontroller (×1)

- **Microcontroller:**ATmega328P
- **Digital I/O Pins:**14 (6 support PWM output)
- **Analog Input Pins:**6 (10-bit ADC, 0–5V range)
- **Operating Voltage:**5V (USB or external supply)
- **Clock Speed:**16 MHz
- **Serial Baud Rate (used):**9600 bps

Engineering Purpose:

The Arduino Uno serves as the central processing unit for the entire monitoring and control subsystem. It reads digital data from the three DHT22 sensors (pins D4, D7, D8) and the float switch (D6), and reads the analog voltage from the MPX5010DP via A0. Based on these inputs it executes the hysteresis-based pump control logic, computes four real-time performance metrics, and outputs all data to the Serial Monitor every 2 seconds. The 2-second polling interval matches the DHT22's minimum sampling requirement. Sensor validation using `isnan()` checks ensures that a failed sensor does not crash the control loop.

2.6 Arduino Control Code — Key Logic

The complete control logic was programmed in Arduino C++ and uploaded to the Uno via USB. Below are the key functional excerpts from the final code:

Pin Definitions & Sensor Objects:

```
#define DHT_PIN_1  4  // DHT22 #1 → Inlet
#define DHT_PIN_2  7  // DHT22 #2 → After desiccant disk
#define DHT_PIN_3  8  // DHT22 #3 → Exit / outlet
#define FLOAT_PIN  6  // Float sensor → water level detection
#define PUMP_RELAY  9  // Relay IN pin → controls pump
#define PRESSURE_PIN A0 // MPX5010DP VOUT
```

Pressure Sensor Transfer Function (MPX5010DP):

The analog voltage from the pressure sensor is converted to kPa using the datasheet transfer function: $V_{out} = V_s \times (0.09 \times P + 0.04)$, rearranged as:

```
float readPressure_kPa() {
  int raw    = analogRead(PRESSURE_PIN);
  float voltage = raw * (5.0 / 1023.0);
  float pressure = ((voltage / 5.0) - 0.04) / 0.09;
  if (pressure < 0.0) pressure = 0.0; // clamp noise
  if (pressure > 10.0) pressure = 10.0; // clamp max
  return pressure;
}
```

Pump Hysteresis Control Logic:

To prevent rapid relay switching (relay chatter), a hysteresis band is implemented: the pump activates when the outlet temperature exceeds 28°C and deactivates only when it drops below 26°C. Between these values the pump holds its previous state.

```
const float PUMP_ON_TEMP = 28.0;
const float PUMP_OFF_TEMP = 26.0;
static bool pumpState = false;

if (!waterPresent) {
  pumpState = false; pumpOFF(); // Tank empty — protect pump
} else if (dht3OK) {
  if (temp3 > PUMP_ON_TEMP) pumpState = true;
  else if (temp3 < PUMP_OFF_TEMP) pumpState = false;
  // Between 26–28 C: hold previous state
}
```

Performance Metrics Computed in Real Time:

After each sensor reading cycle, the Arduino computes and prints the following metrics to the Serial Monitor at 9600 baud:

- Temperature Reduction: Inlet temp minus outlet temp (°C) — quantifies overall cooling achieved
- Humidity Rise across evaporative section: Outlet RH minus inlet RH (%) — confirms evaporation is occurring
- Desiccant Effect: Inlet RH minus post-desiccant RH (%) — confirms the desiccant disc is actively removing moisture
- Pressure Drop: Real-time differential pressure across the desiccant disc (kPa and mbar)

All readings are refreshed every 2 seconds, which matches the DHT22's minimum polling interval. Sensor validation checks (isnan() guards) ensure that a failed sensor does not crash the control loop — instead, the pump defaults to safe ON mode and an error flag is printed to Serial.

2.7 4-Channel Relay Module (×1)

- **Type:** Opto-isolated 4-channel relay board
- **Relay Rating:** 10A / 250V AC or 10A / 30V DC per channel
- **Trigger:** Active HIGH (D9 HIGH → pump ON | D9 LOW → pump OFF)

Engineering Purpose:

The Arduino's digital output pins source only 5V at 40mA — far below what the submersible water pump requires. The relay module bridges this gap: Channel 1 is connected to D9 and controls the pump. When the Arduino calls pumpON(), it drives D9 HIGH, closing the relay contact and completing the pump's power circuit. When pumpOFF() is called, D9 goes LOW and the relay opens, cutting power to the pump. The opto-isolation ensures that any back-EMF or voltage spike from the pump motor cannot damage the Arduino's GPIO.

2.8 YF-S201 Water Flow Sensor (×1)

- **Measurement Principle:** Hall-effect turbine pulse counting
- **Flow Rate Range:** 1–30 L/min
- **Output:** Digital pulse signal; pulse frequency proportional to flow rate

Engineering Purpose:

The YF-S201 is installed inline in the water supply line from the submersible pump to the upper distribution pipe of the honeycomb pads. It measures the actual volumetric flow rate of water reaching the evaporative pads. If the pump is running but flow drops — indicating a blocked line or failed pump — the fault can be identified. It also provides water consumption data that complements the thermodynamic readings from the DHT22 sensors.

3. Fabrication, Procurement, and Cost Analysis

3.1 Electronic Components — Procurement Log & Cost

All electronic components were sourced from local electronics markets in Lahore. The following table documents the complete procurement log for our group's scope:

#	Component	Qty	Unit Cost (PKR)	Total (PKR)	Source
1	DHT22 Temp+Humidity Sensor	3	375	1,125	Local Electronics Market
2	DS18B20 Temperature Sensor	2	225	450	Local Electronics Market
3	MPX5010DP Pressure Sensor	1	500	500	Industrial Electronics Sourcing
4	Float Switch (Water Level)	1	125	125	Local Hardware Supplier
5	Arduino Uno Microcontroller	1	1,350	1,350	Local Electronics Market
6	4-Channel Relay Module	1	450	450	Local Electronics Market
7	PWM Fan Speed Controller	1	350	350	Local Electronics Market
8	YF-S201 Water Flow Sensor	1	300	300	Local Electronics Market
9	Jumper Wires (pack)	1 pack	150	150	Local Electronics Market
10	Breadboard	1	225	225	Local Electronics Market
11	5V 2A Power Adapter	1	450	450	Local Electronics Market
12	4.7kΩ Resistors	2	15	30	Local Electronics Market
13	USB Cable (Arduino)	1	175	175	Local Market

Total Electronic Components Investment	≈ 7,480 PKR
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3.2 Fitting & Mounting Materials — Procurement Log & Cost

In addition to electronic components, specialized mounting hardware, sealing materials, and fittings were required for the physical installation of all sensors. These were sourced from local hardware and general markets:

#	Item	Use	Qty	Price (PKR)	Total (PKR)
1	Hot glue gun	Mounting & sealing sensors	1	Rs 300–400	Rs 350
2	Hot glue sticks (pack)	Refills for glue gun	1 pack	Rs 100–150	Rs 120

#	Item	Use	Qty	Price (PKR)	Total (PKR)
3	Zip ties pack	Securing wires and sensors	1 pack	Rs 100–150	Rs 120
4	Electrical tape	Insulating wire joints	1 roll	Rs 50–80	Rs 60
5	Waterproof tape	Float switch wire connections	1 roll	Rs 80–100	Rs 80
6	Aquarium air tube (1m)	Pressure sensor tubing	1m	Rs 50–100	Rs 70
7	Teflon tape	Sealing water pipe fittings	1 roll	Rs 30–50	Rs 40
8	Hose clamps / jubilee clips	Securing YF-S201 on pipe	2	Rs 50–80	Rs 130
9	Aluminum tape	Holding DS18B20 on hot surface	1 roll	Rs 100–150	Rs 125
10	Small plastic container / box	Control box for Arduino & electronics	1	Rs 150–250	Rs 200
11	Double sided tape	Mounting Arduino & relay inside box	1 roll	Rs 80–100	Rs 90
12	Small screws + nuts (pack)	Extra securing of components	1 pack	Rs 50–80	Rs 65
13	Sandpaper (small piece)	Roughen surface before gluing	1 piece	Rs 20–30	Rs 25
14	Marker / label stickers	Label each sensor point	1	Rs 50–80	Rs 65

Total Fitting & Mounting Hardware Investment	≈ 1,340 PKR
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TOTAL GROUP PROCUREMENT INVESTMENT	≈ 8,820 PKR
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3.3 Arduino Pin Assignment Summary

The following table summarizes the complete pin mapping between the Arduino Uno and all connected sensors and actuators:

Arduino Pin	Connected To	Function
D4	DHT22 #1 (Inlet)	Digital data signal from inlet temperature & humidity sensor
D7	DHT22 #2 (Post-Desiccant)	Digital data signal from sensor after desiccant disc

Arduino Pin	Connected To	Function
D8	DHT22 #3 (Outlet / Exit)	Digital data signal from cooled air outlet sensor
D6	Float Switch	Water level detection (INPUT_PULLUP; LOW = water present)
D9	Pump Relay (IN)	Controls submersible water pump ON/OFF via relay module
A0	MPX5010DP VOUT	Analog voltage from differential pressure sensor (0.2V–4.7V)

3.4 Sensor Installation & Wiring Workflow

To ensure reliable readings and safe operation, the team followed a structured four-stage installation process:

3. **Sensor Placement & Mechanical Mounting:** All sensors were positioned at designated measurement points on the cooler casing. DHT22 sensors were secured using hot glue at the inlet grille (D4), post-desiccant duct (D7), and outlet duct (D8). DS18B20 probes were fixed to the regeneration duct using aluminum tape to maintain contact with the hot surface. The MPX5010DP was mounted externally on the control box, with aquarium air tubes routed through Teflon-sealed fittings to the pressure tap points on either side of the desiccant disc.
4. **Wiring & Circuit Assembly:** All sensor signal wires were routed along the interior casing wall and secured with zip ties to keep them away from moving fan blades and water lines. A breadboard was used for all connections to allow easy debugging and modification. Pull-up resistors for the DS18B20 and DHT22 sensors were installed on the breadboard. The float switch was wired to D6 with the INPUT_PULLUP configuration, eliminating the need for an external resistor.
5. **Arduino Programming & Relay Integration:** The Arduino was programmed using the DHT library by Adafruit and DallasTemperature library. The control loop runs every 2 seconds, reads all sensors, computes performance metrics, controls the pump relay, and outputs all data to Serial at 9600 baud. Each relay channel was tested individually before full system integration.
6. **System Verification & Calibration:** All sensors were powered and readings verified on the Serial Monitor before the system was sealed. The float switch was tested by manually lowering the water level to confirm the pump shutoff response. The pressure sensor output was verified by partially restricting airflow across the desiccant disc and observing the rising kPa reading on Serial.

4. Engineering Challenges Faced & Corrective Actions

During sensor installation and system integration, the team encountered several practical engineering challenges. The following documents the problems and solutions:

4.1 DHT22 Sensor Placement — Thermal Cross-Contamination

- **The Challenge:**

Initial placement of the post-desiccant DHT22 sensor (D7) was too close to the regeneration duct partition wall. Heat conducted through the polymer partition elevated the local temperature at the sensor, producing readings that falsely indicated a larger temperature rise than what the desiccant wheel alone was producing. This would have overstated the desiccant effect metric.

- **Corrective Action:**

The sensor was repositioned to the geometric center of the post-desiccant air channel, mounted on a small standoff made from hot-glued acrylic strip to prevent conductive heat transfer from the surface. Readings stabilized and became consistent with expected desiccant outlet conditions.

4.2 Pressure Sensor Tubing Leakage

- **The Challenge:**

The aquarium air tubes connecting the MPX5010DP ports to the desiccant disc pressure taps were initially slip-fitted without any sealant. At operating fan speeds, the negative pressure inside the cooler caused the soft tubing to partially collapse at bends, producing erratic and non-repeatable pressure readings.

- **Corrective Action:**

Teflon tape was applied at all tube-to-fitting connections to ensure an airtight seal, and the tube routing was redesigned to eliminate sharp bends. Hose clamps were added at the barbed fittings of the sensor to mechanically lock the tubing in place. After these changes, pressure readings became stable and repeatable.

4.3 Pump Relay Chattering at Threshold Temperature

- **The Challenge:**

During early testing, when the outlet temperature hovered near the control threshold, the relay was switching ON and OFF rapidly — multiple times per second — causing audible chattering and risking relay contact wear. This happened because the initial code used a single threshold value, so any minor sensor noise above or below it triggered a state change.

- **Corrective Action:**

A hysteresis band was implemented in the control code: the pump turns ON only when outlet temperature exceeds 28°C, and turns OFF only when it drops below 26°C. Within this 2°C band, the pump holds its previous state. This completely eliminated relay chattering and stabilized pump operation.

5. Conclusion & Recommendations

5.1 Project Conclusion

The sensor and control group successfully designed, procured, and integrated a complete electronic monitoring and automation layer for the desiccant-enhanced evaporative water cooler prototype. The system provides full real-time observability of the thermodynamic process through strategically positioned temperature, humidity, pressure, and flow sensors. The Arduino-based control logic enables autonomous operation — including pump protection via float switch and temperature hysteresis control — without requiring manual intervention during demonstration. All sensor readings and four computed performance metrics are output to the Serial Monitor in real time, providing the thermodynamic evidence required for the OEL assessment.

All components were sourced within budget constraints using locally available parts from Lahore's electronics markets. The combined group investment of approximately 8,820 PKR delivered a functional, instrumented, and automated control subsystem.

5.2 Recommendations for Future Iterations

- **Data Logging via SD Card Module:** Adding an SD card module to the Arduino would allow continuous logging of all sensor readings and computed metrics to a CSV file, enabling proper time-series analysis of cooling and regeneration cycles.
- **LCD Display Integration:** A 16×2 I2C LCD display could be added to show live temperature and humidity readings directly on the prototype without requiring a laptop connection to the Serial Monitor, improving usability during demonstration.
- **Wireless Monitoring via ESP8266/ESP32:** Replacing or supplementing the Arduino Uno with an ESP-based board would enable Wi-Fi connectivity, allowing sensor data to be streamed to a mobile device in real time.
- **PID Control for Pump Speed:** Rather than binary ON/OFF relay control, a PWM-controlled pump driver with a PID loop could modulate water flow rate proportionally to the outlet temperature, providing more precise and energy-efficient evaporative cooling.